Degradation under Gaussian weight perturbation: muon vs adamw
chinchilla (token budget)
model size
muon - adamw loss degradation
0.4
0.2
0.0
−0.2
−0.4

$\gamma = 0.003$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0005			
64	+0.0010				
32	+0.0006	+0.0004	+0.0004		
16	+0.0003	+0.0002	+0.0004		
8	+0.0002	+0.0002	+0.0003	+0.0000	
4	+0.0001	+0.0002	+0.0001	+0.0000	
2	+0.0001	+0.0001	+0.0001	+0.0001	+0.0001
1	+0.0001	+0.0001	+0.0001	+0.0000	+0.0000
0.5	+0.0001	+0.0000	-0.0000	+0.0000	+0.0000
0.25	-0.0000	+0.0001	+0.0001	-0.0000	+0.0000

$\gamma = 0.005$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0013			
64	+0.0027				
32	+0.0016	+0.0009	+0.0011		
16	+0.0010	+0.0006	+0.0011		
8	+0.0008	+0.0005	+0.0007	+0.0001	
4	+0.0004	+0.0006	+0.0002	+0.0001	
2	+0.0004	+0.0003	+0.0002	+0.0001	+0.0002
1	+0.0003	+0.0001	+0.0002	+0.0001	+0.0001
0.5	+0.0002	+0.0001	+0.0000	+0.0000	+0.0000
0.25	+0.0001	+0.0001	+0.0001	+0.0000	+0.0000

$\gamma = 0.007$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0026			
64	+0.0049				
32	+0.0029	+0.0016	+0.0022		
16	+0.0018	+0.0011	+0.0020		
8	+0.0015	+0.0010	+0.0012	+0.0002	
4	+0.0006	+0.0011	+0.0004	+0.0001	
2	+0.0007	+0.0005	+0.0004	+0.0002	+0.0003
1	+0.0005	+0.0003	+0.0003	+0.0001	+0.0001
0.5	+0.0004	+0.0002	+0.0001	+0.0001	+0.0001
0.25	-0.0001	+0.0003	+0.0002	+0.0001	+0.0001

$\gamma = 0.01$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0052			
64	+0.0091				
32	+0.0058	+0.0033	+0.0044		
16	+0.0036	+0.0023	+0.0038		
8	+0.0030	+0.0020	+0.0025	+0.0003	
4	+0.0014	+0.0022	+0.0009	+0.0003	
2	+0.0014	+0.0009	+0.0009	+0.0005	+0.0006
1	+0.0010	+0.0006	+0.0006	+0.0003	+0.0003
0.5	+0.0006	+0.0004	+0.0002	+0.0002	+0.0002
0.25	+0.0002	+0.0005	+0.0003	+0.0001	+0.0001

$\gamma = 0.02$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0211			
64	+0.0380				
32	+0.0233	+0.0127	+0.0183		
16	+0.0150	+0.0094	+0.0149		
8	+0.0119	+0.0078	+0.0096	+0.0011	
4	+0.0058	+0.0084	+0.0037	+0.0013	
2	+0.0059	+0.0035	+0.0034	+0.0017	+0.0023
1	+0.0043	+0.0024	+0.0025	+0.0009	+0.0011
0.5	+0.0030	+0.0018	+0.0012	+0.0008	+0.0008
0.25	+0.0012	+0.0016	+0.0011	+0.0004	+0.0004

$\gamma = 0.03$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0498			
64	+0.0850				
32	+0.0523	+0.0293	+0.0423		
16	+0.0336	+0.0217	+0.0351		
8	+0.0271	+0.0177	+0.0218	+0.0019	
4	+0.0134	+0.0191	+0.0084	+0.0029	
2	+0.0134	+0.0079	+0.0080	+0.0037	+0.0050
1	+0.0096	+0.0053	+0.0055	+0.0019	+0.0027
0.5	+0.0069	+0.0042	+0.0026	+0.0018	+0.0018
0.25	+0.0031	+0.0031	+0.0025	+0.0009	+0.0010

$\gamma = 0.04$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0934			
64	+0.1549				
32	+0.0932	+0.0528	+0.0782		
16	+0.0614	+0.0398	+0.0645		
8	+0.0483	+0.0324	+0.0407	+0.0033	
4	+0.0244	+0.0345	+0.0153	+0.0062	
2	+0.0245	+0.0141	+0.0144	+0.0066	+0.0094
1	+0.0169	+0.0097	+0.0102	+0.0033	+0.0054
0.5	+0.0123	+0.0076	+0.0048	+0.0031	+0.0036
0.25	+0.0058	+0.0053	+0.0045	+0.0016	+0.0020

$\gamma = 0.05$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.1555			
64	+0.2472				
32	+0.1503	+0.0852	+0.1273		
16	+0.0972	+0.0638	+0.1049		
8	+0.0750	+0.0519	+0.0649	+0.0055	
4	+0.0394	+0.0547	+0.0243	+0.0099	
2	+0.0388	+0.0222	+0.0232	+0.0107	+0.0149
1	+0.0267	+0.0154	+0.0162	+0.0052	+0.0084
0.5	+0.0194	+0.0120	+0.0076	+0.0053	+0.0054
0.25	+0.0095	+0.0080	+0.0071	+0.0024	+0.0033

$\gamma = 0.07$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.3499			
64	+0.4923				
32	+0.2991	+0.1789	+0.2792		
16	+0.1930	+0.1348	+0.2283		
8	+0.1488	+0.1096	+0.1375	+0.0114	
4	+0.0812	+0.1128	+0.0512	+0.0245	
2	+0.0766	+0.0444	+0.0482	+0.0223	+0.0330
1	+0.0525	+0.0322	+0.0337	+0.0103	+0.0182
0.5	+0.0381	+0.0246	+0.0157	+0.0109	+0.0118
0.25	+0.0191	+0.0156	+0.0143	+0.0049	+0.0069

$\gamma = 0.1$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.8200			
64	+0.9587				
32	+0.6282	+0.4067	+0.6876		
16	+0.3947	+0.3072	+0.5550		
8	+0.3067	+0.2499	+0.3225	+0.0294	
4	+0.1777	+0.2558	+0.1200	+0.0806	
2	+0.1644	+0.0966	+0.1100	+0.0519	+0.0830
1	+0.1079	+0.0754	+0.0775	+0.0230	+0.0470
0.5	+0.0784	+0.0547	+0.0346	+0.0247	+0.0276
0.25	+0.0401	+0.0322	+0.0307	+0.0109	+0.0147

$\gamma = 0.13$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+1.3538			
64	+1.4211				
32	+1.0443	+0.7543	+1.3706		
16	+0.6153	+0.5727	+1.0814		
8	+0.5130	+0.4585	+0.6458	+0.0518	
4	+0.3187	+0.4809	+0.2335	+0.2422	
2	+0.2928	+0.1746	+0.2149	+0.1041	+0.1779
1	+0.1843	+0.1492	+0.1528	+0.0446	+0.1040
0.5	+0.1338	+0.1035	+0.0646	+0.0486	+0.0551
0.25	+0.0710	+0.0565	+0.0560	+0.0209	+0.0288

cell = (perturbed - unperturbed) for muon minus the same for adamw | negative (blue) = muon degrades less | white = no data

colour scale clipped at ±0.579 (97th pct); 12 cell(s) beyond it keep their printed value